2024 MIT Science Bowl

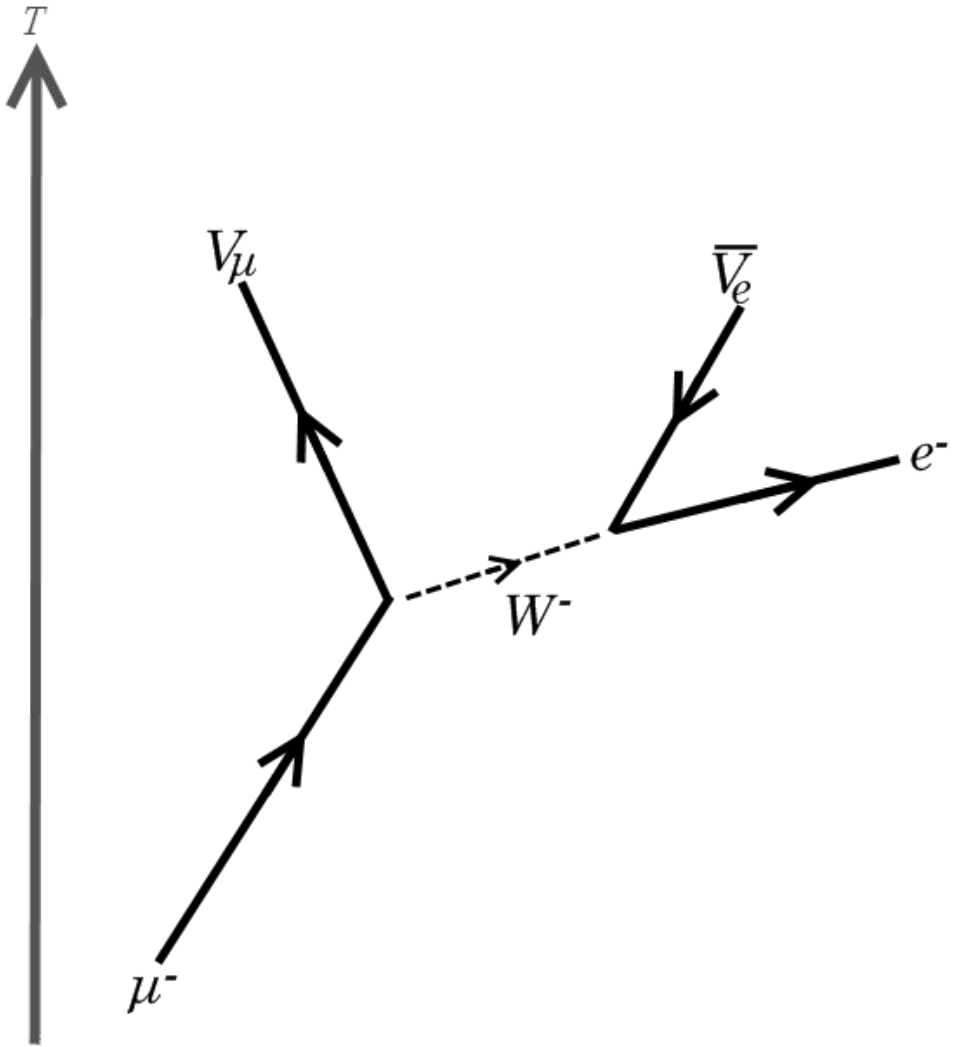
Round 12



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Bonus 1 (Physics)

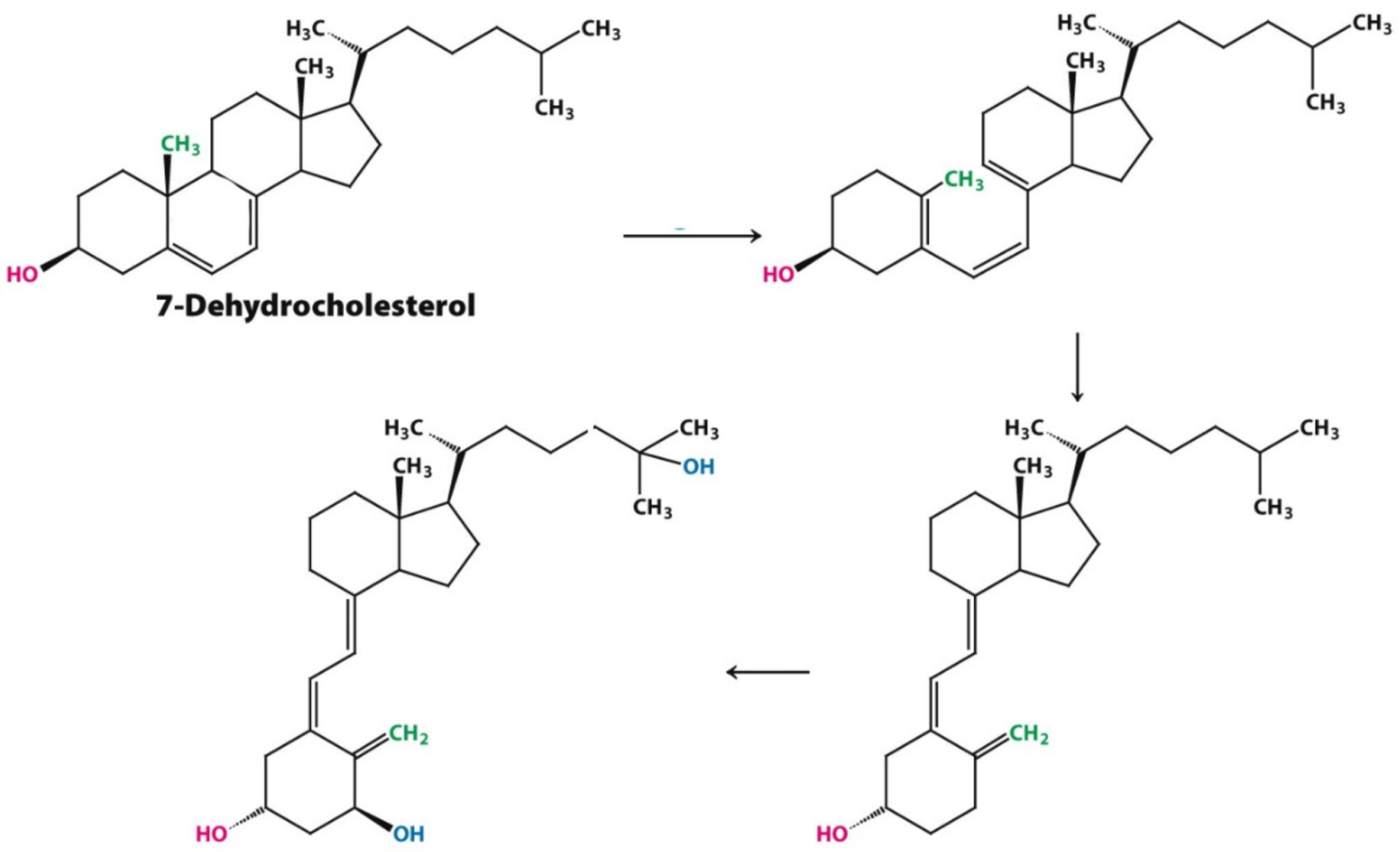

T
ν_μ
$\overline{\nu}_e$
e^-
W^-
μ^-



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Bonus 5 (Chem)

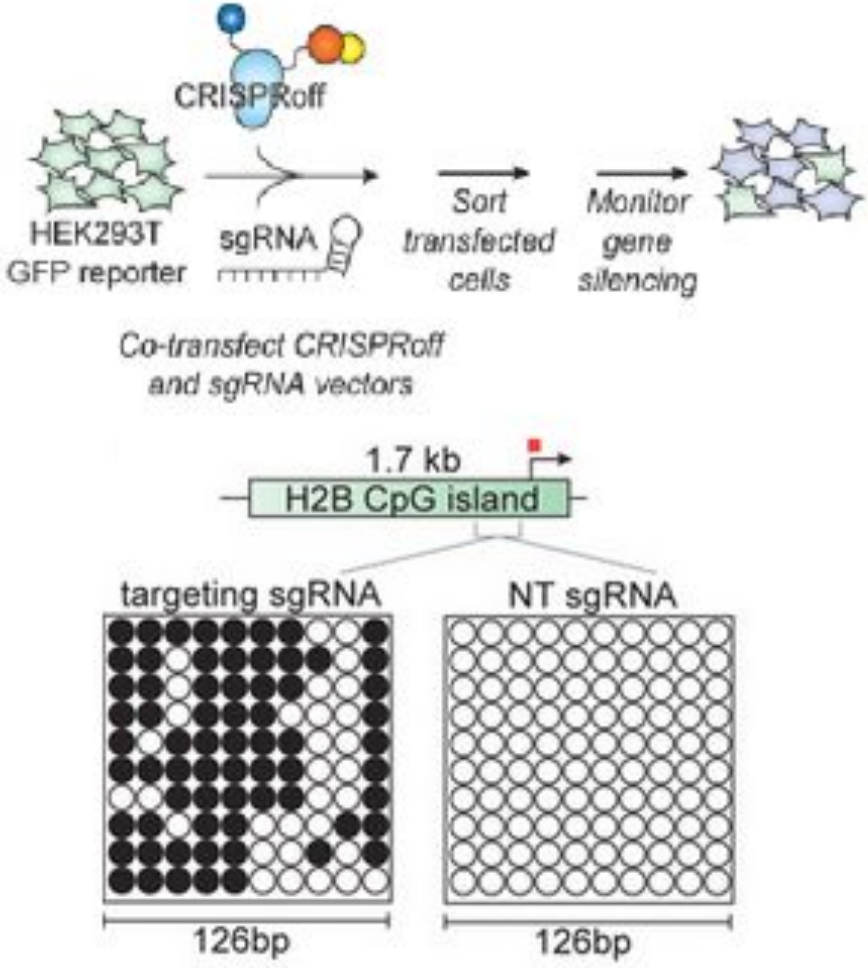

H3C
CH3
CH3
CH3
CH3
HO
7-Dehydrocholesterol
CH3
CH3
CH3
CH3
HO
CH3
CH3
CH3
CH3
HO
CH2
CH3
CH3
CH3
OH
HO
CH2
OH



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Bonus 7 (Energy)


CRISPRoff
HEK293T
GFP reporter
sgRNA
Sort
transfected
cells
Monitor
gene
silencing
Co-transfect CRISPRoff
and sgRNA vectors
1.7 kb
H2B CpG island
targeting sgRNA
NT sgRNA
126bp
126bp

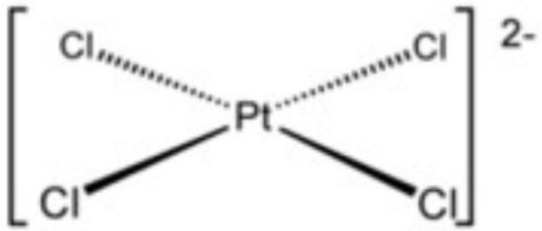


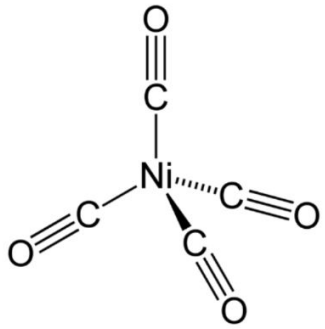
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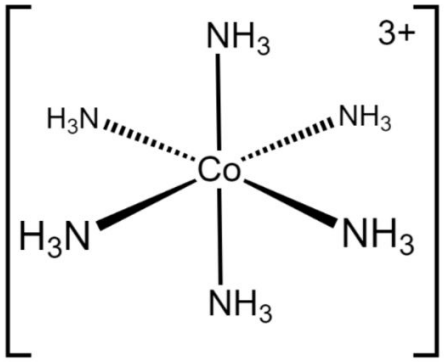
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Bonus 8 (Chem)


E
A
1
B
2
C
3


Cl
Cl
Pt
Cl
Cl
2-


O
C
Ni
C
O
C
O
C
O

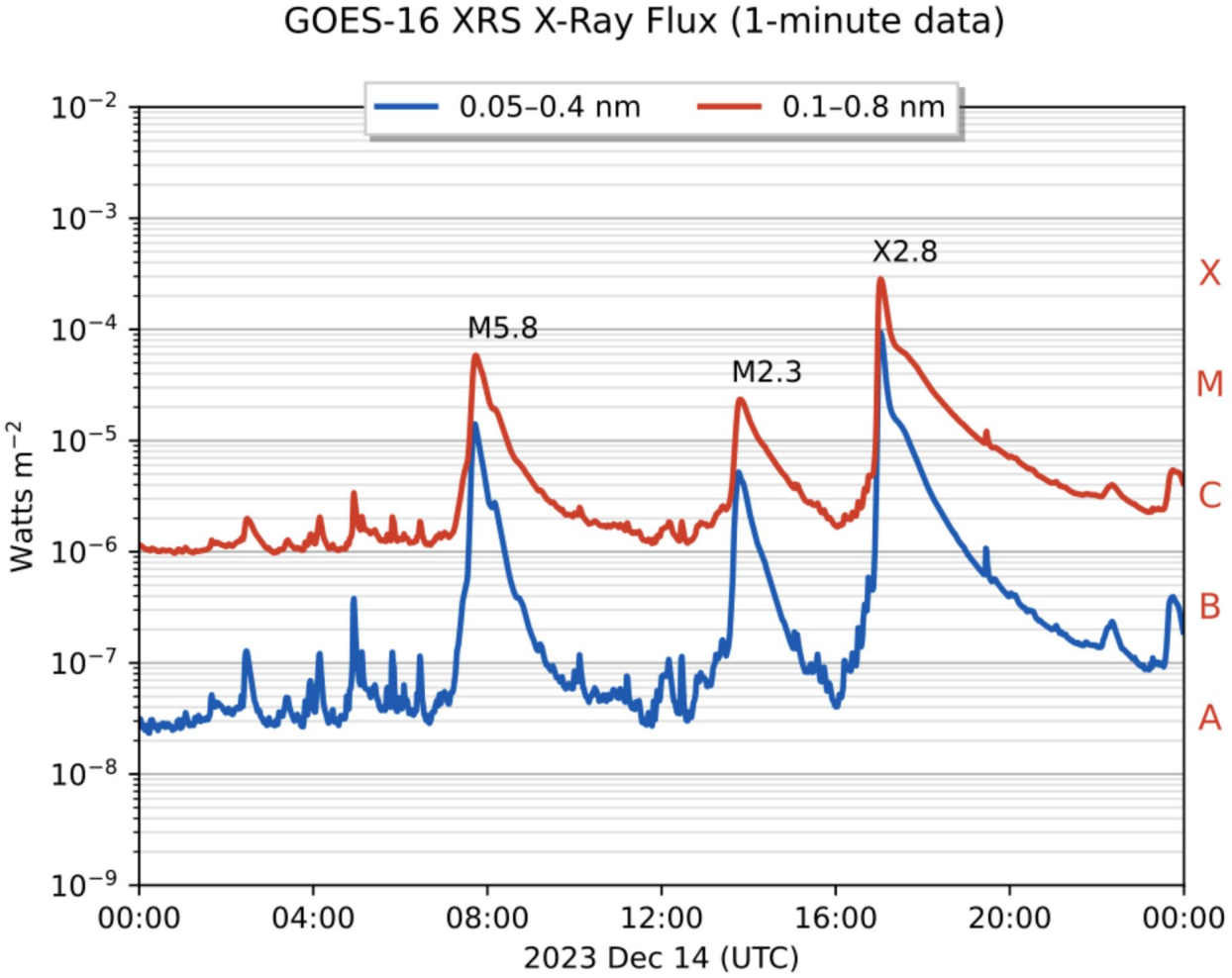

NH3
3+
H3N
NH3
Co
H3N
NH3
NH3



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Bonus 12 (ESS)

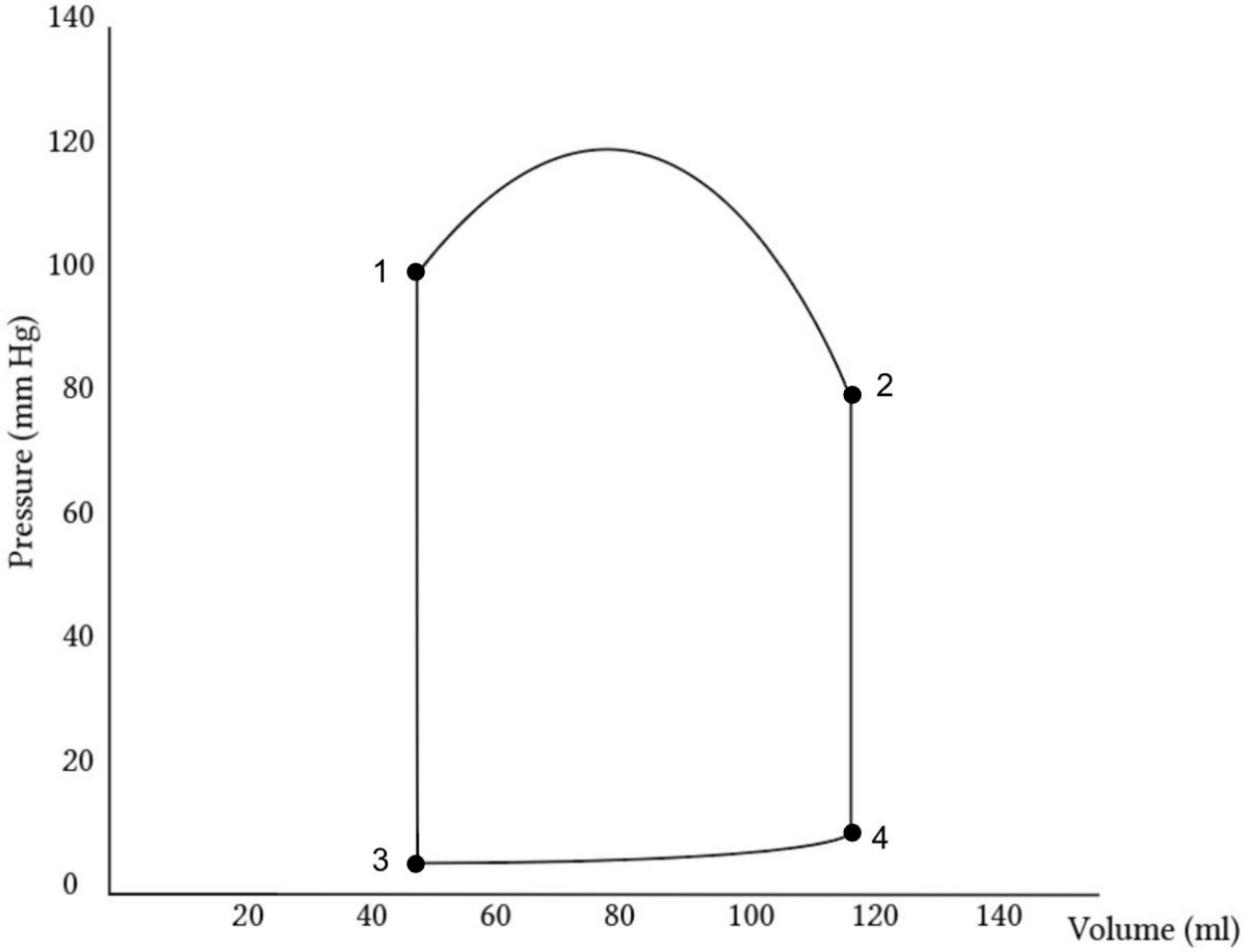

GOES-16 XRS X-Ray Flux (1-minute data)
0.05–0.4 nm
0.1–0.8 nm
M5.8
M2.3
X2.8
X
M
C
B
A
Watts m^{-2}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
00:00
04:00
08:00
12:00
16:00
20:00
00:00
2023 Dec 14 (UTC)



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Bonus 15 (Bio)

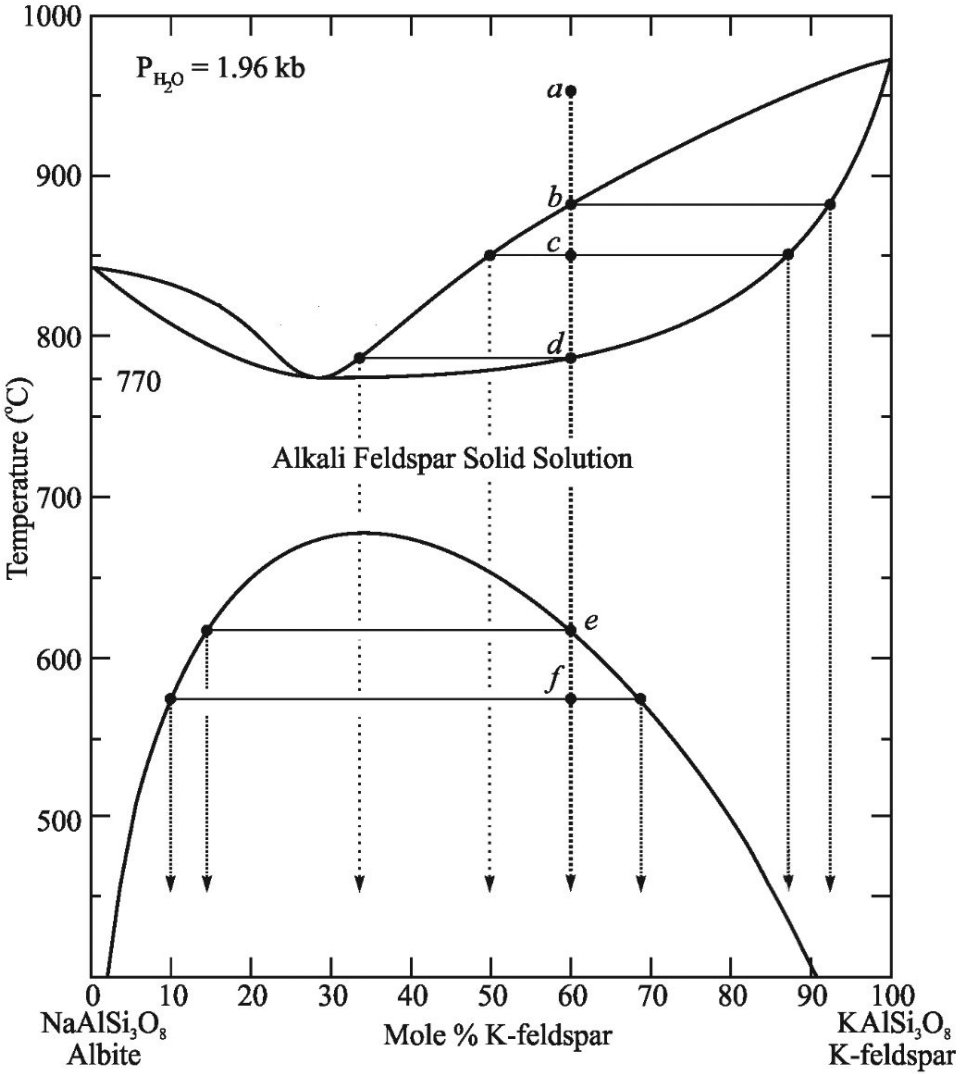

140
120
100
80
60
40
20
0
Pressure (mm Hg)
1
2
3
4
20
40
60
80
100
120
140
Volume (ml)



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Bonus 17 (ESS)


1000
900
800
700
600
500
Temperature (°C)
P_{H_2O} = 1.96 kb
a
b
c
d
770
Alkali Feldspar Solid Solution
e
f
0
10
20
30
40
50
60
70
80
90
100
$NaAlSi_3O_8$
Albite
Mole % K-feldspar
$KAlSi_3O_8$
K-feldspar



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Bonus 21 (Bio)



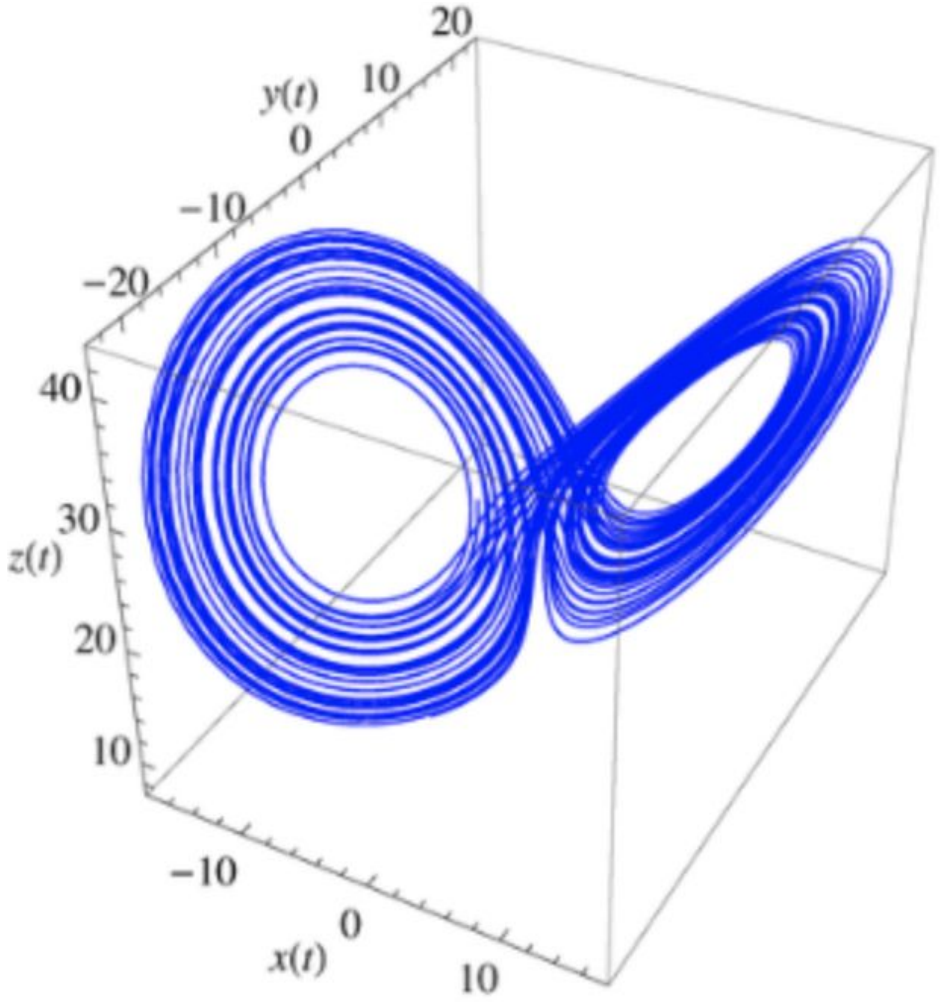
Image Credit: Rasbak, Wikimedia Commons



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Bonus 22 (Math)


20
10
y(t)
0
−10
−20
40
30
z(t)
20
10
−10
0
x(t)
10



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